



UNIVERSITATEA
DIN BUCUREȘTI

Metode de dezvoltare software

Organizare curs

24.02.2026

Metode de dezvoltare software - în practică...



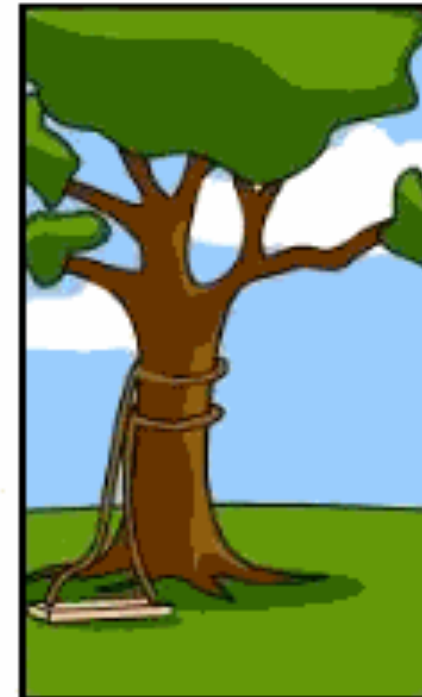
How the customer explained it



How the Project Leader understood it



How the Analyst designed it



How the Programmer wrote it



How the Business Consultant described it



How the project was documented



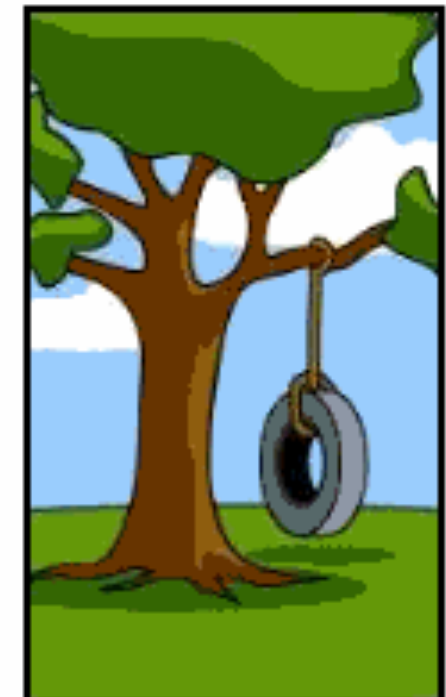
What operations installed



How the customer was billed

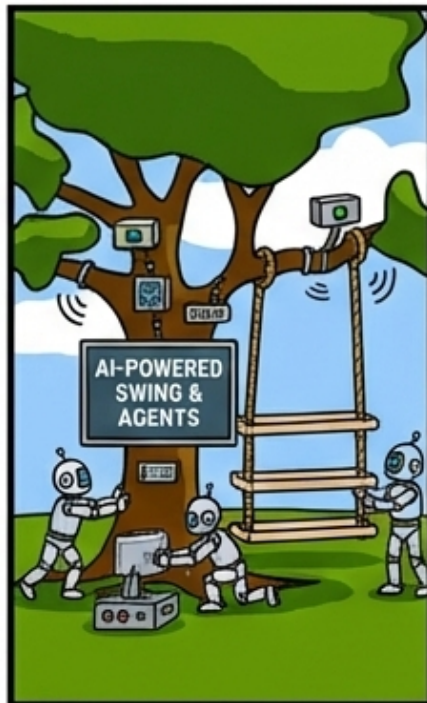


How it was supported



What the customer really needed

Metode de dezvoltare software cu Agentic AI - în practică...



How the customer explained it with AI.



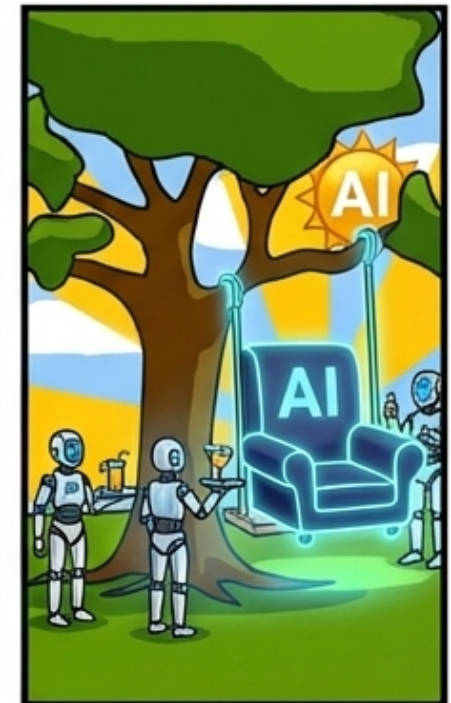
How the Project Leader understood the AI Agents.



How the Analyst designed the Agentic AI.



How the Programmer wrote the AI Agent code.



How the Business Consultant described the Agentic AI.



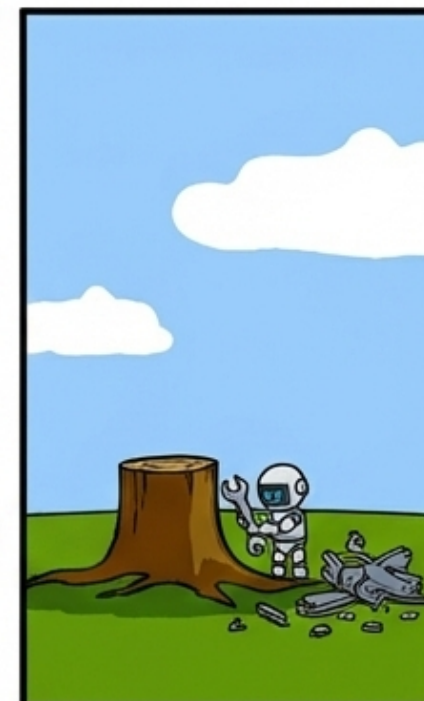
How the AI project was documented.



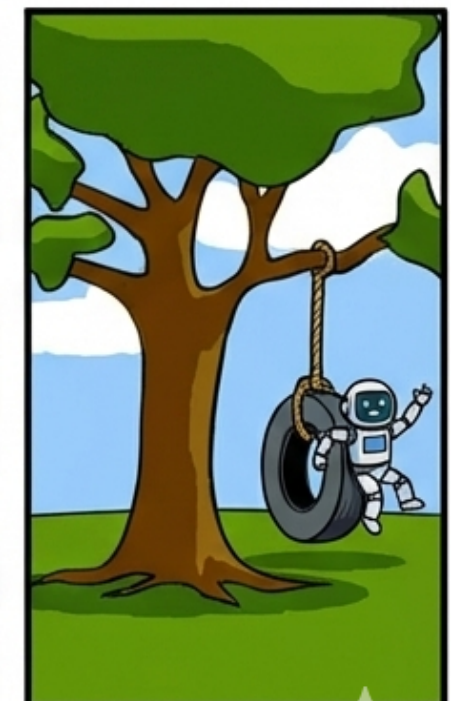
What operations installed with AI Agents.



How the customer was billed for Agentic AI.



How the AI was supported.



What the customer really needed from AI.

Scurt bio - Alin Ștefănescu

Licență & Master (1994-2000) în informatică la **Univ. din București**

Doctorat (2000-2005) în informatică la **Univ. din Stuttgart**, Germania
cu stagii in Edinburgh, Scoția și München, Germania.

Postdoc (2005-2006) la **Univ. din Konstanz**, Germania.

Cercetător (2007-2010) la **SAP Research**, Germania.

Manager (2010-prezent) proiecte R&D cu industria (UiPath, Bitdefender, Adobe etc.)

Co-fondator (2022-prezent) la **Institute for Logic and Data Science**

Profesor (2013-prezent) la **Univ. din București**

Evangelist Agentic AI (2026-prezent) la **cursul MDS**

Organizare curs



programmare

CURS
MDS

aspecte care
faciliteaza
programarea

agentic software engineering

agentic AI for
software engineering

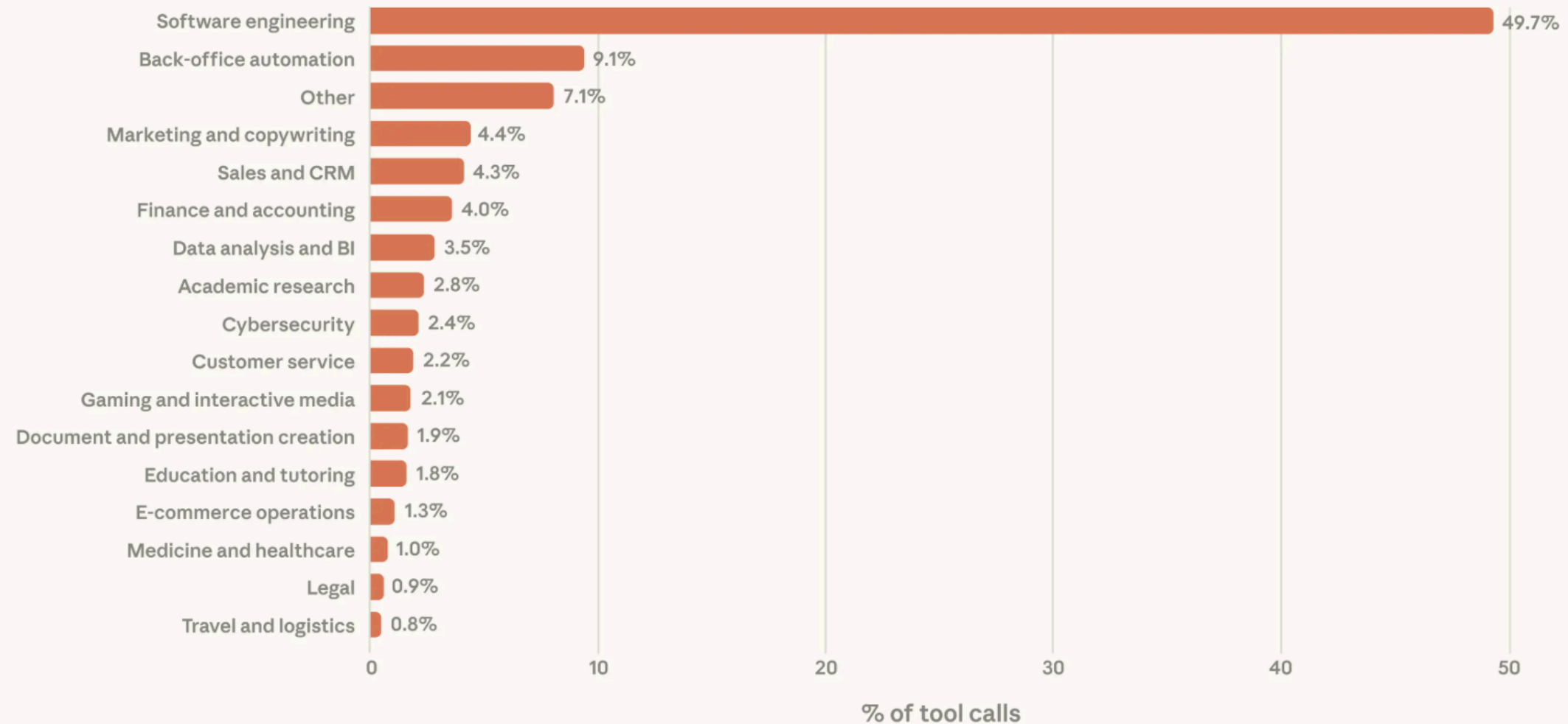
A Venn diagram with two overlapping circles. The top circle is green and contains the text 'agentic AI for software engineering'. The bottom circle is yellow and contains the text 'software engineering for AI agents'. The intersection of the two circles is shaded in a darker green.

software
engineering
for AI agents

CURS
MDS

Unde sunt agenții folosiți acum

In what domains are agents deployed?



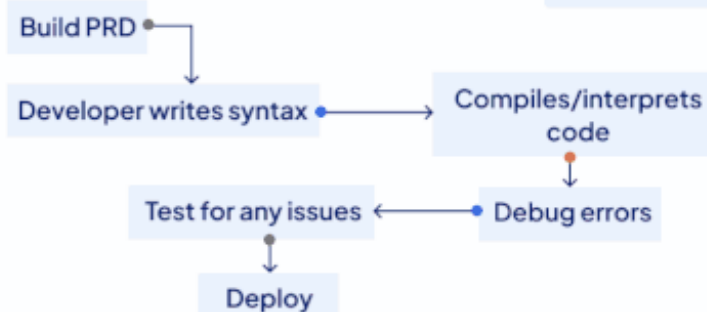
Statistici Anthropic de la

<https://www.anthropic.com/research/measuring-agent-autonomy>

Evolution of Coding with AI Agents

Right methods from development to Deployment

Traditional Coding



What exactly is traditional coding?

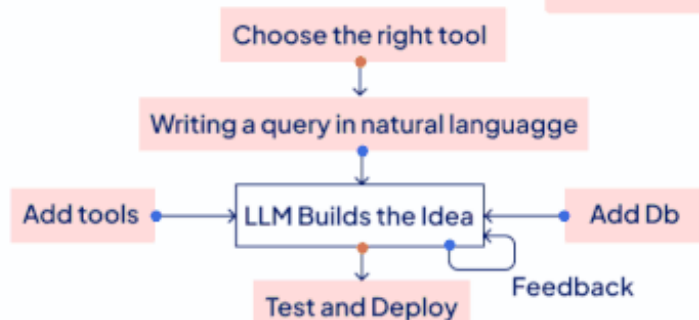
Writing code manually line-by-line in a programming language

When to use traditional coding?

When you need full control, custom logic, or complex architecture

Tools to use VsCode IntelliJ Sublime text

Vibe coding



What exactly is Vibe Coding?

Describing in plain language and letting AI generate the entire app

When to use Vibe coding?

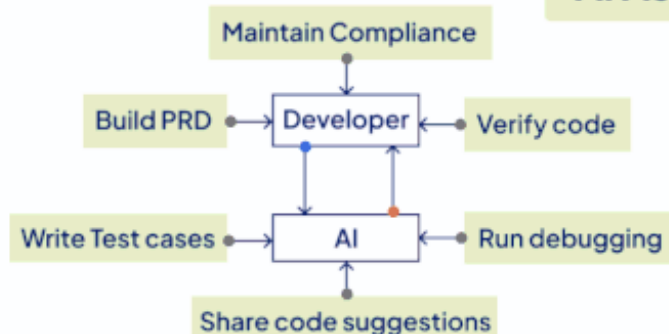
For quick prototypes, simple apps, or learning new frameworks

Tools to use **bolt.new** **Lovable** Replit agent



@rakeshgohe101

AI Assisted Coding



What exactly is AI Assisted coding?

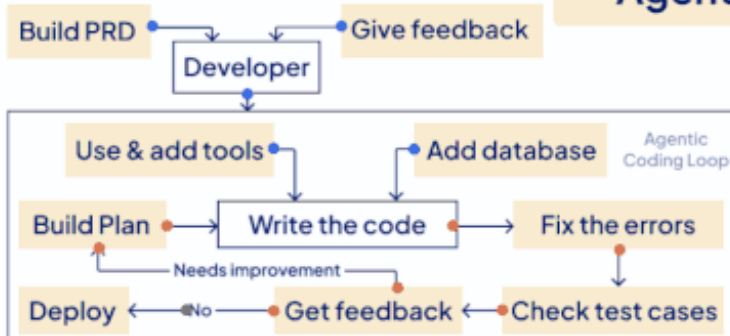
Developer writes code while AI suggests completions

When to use AI Assisted Coding?

For production-grade projects requiring oversight

Tools to use Github Copilot Code Whisperer

Agentic Coding



What exactly is traditional coding?

AI agents autonomously code in iterative loops with developer

When to use traditional coding?

For complex workflows, or end-to-end automation.

Tools to use **CLAUDE CODE** OpenAI Codex



LLM = Brain

- Generates text using reasoning
- Relies only on what it was trained on.



RAG = Brain + Books

- LLM + external knowledge
- Pulls information from documents, databases, and knowledge bases



AI Agent = Brain + Hands

- Can take actions
- Uses tools, memory, and decision-making to execute tasks.



MCP = Nervous System

- Connects everything together
- Enables seamless communication between models, tools, APIs, and systems



Cerinte

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diagrame

Design

refactorizare

procese de dezvoltare

Implementare

build

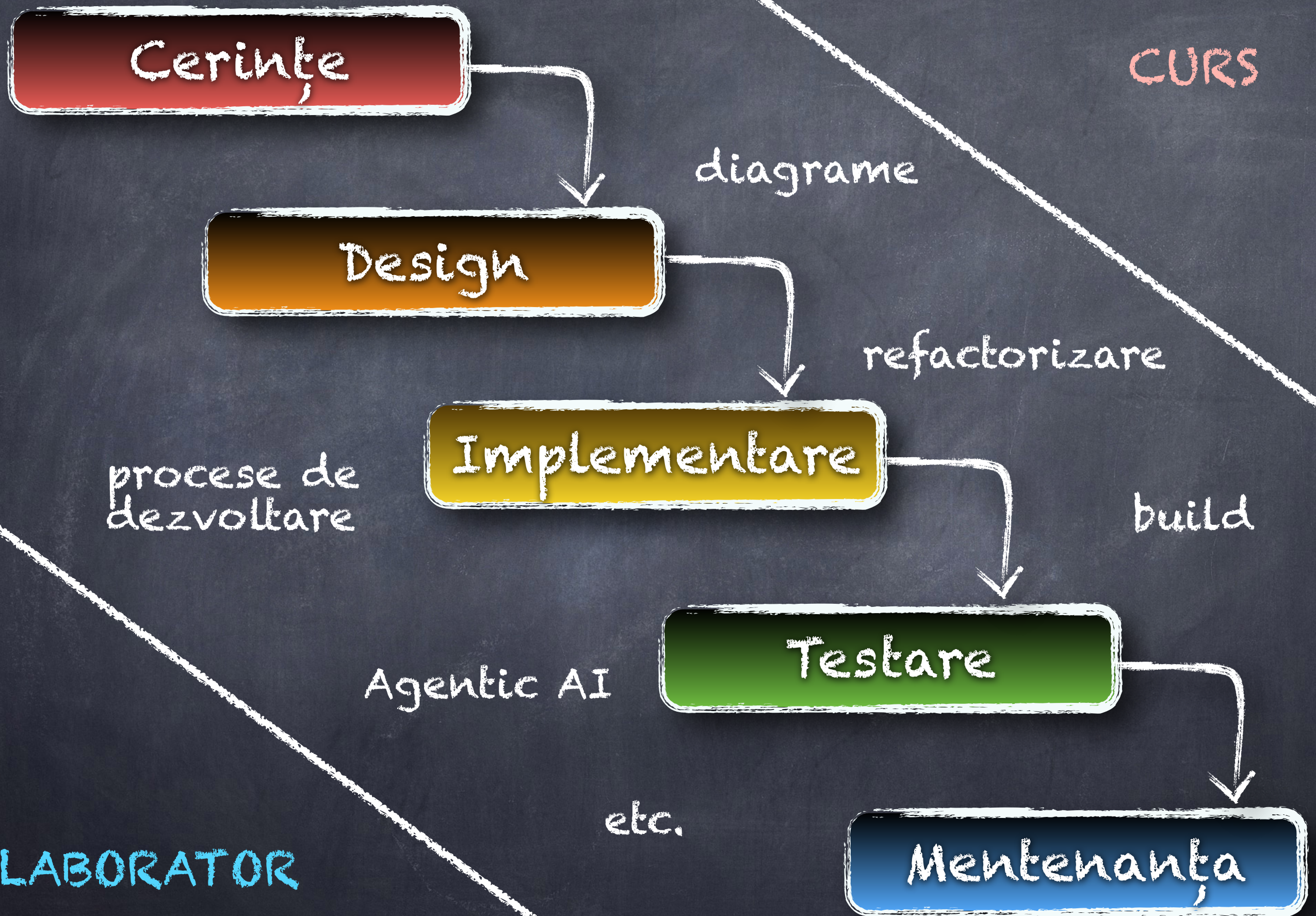
Agentic AI

Testare

etc.

LABORATOR

Mentenanța



Objective

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Descrierea unor
metode practice
de dezvoltare de
produse software

Dezvoltarea în echipa
a unui prototip software,
trecând prin principalele
faze de dezvoltare
folosind Agentic AI

LABORATOR

LABORATOR

Lucru în echipe de 3-5
studenți

Proiect:

1. implementare
(programare în limbajul
favorit)
2. documentarea
procesului

Nota la Laborator:
media celor doua aspecte

CURS

Nota finala:

- 80% Laborator
- 10% referat individual
- 10% punctul din oficiu

Un cavaler nu poate învinge singur!



camarazi de nadejde

o hartă precisă

scut de calitate

strategie de luptă

arme bune

la nevoie, un cal rapid si un AI bun...

Detalii despre proiect și referat

https://is.gd/barem_Lab_MDS | https://is.gd/referat_MDS



Contact

- alin.stefanescu@unibuc.ro

- Course etiquette:



așa NU



așa DA



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- Software Engineering at Google, 2020
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Bibliografie - Agentic Software Engineering

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<https://agenticse-book.github.io>
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<https://themodernsoftware.dev>
- Anthropic Courses on Claude, 2025
<https://anthropic.skilljar.com>
- Andrei Oros. AI-Assisted Software Developement, 2026
<https://abqinstitute.substack.com/p/course-ai-assisted-software-development>
- Microsoft, 2026:
<https://github.com/microsoft/ai-agents-for-beginners>

Promovare oportunități

- Innovation Labs - <https://www.innovationlabs.ro>
- Cariere 15.0 - <https://carriere.asmi.ro>
- Idee: pentru proiect luati un anunt de job si incercati sa atingeti in proiect toate tehnologiile

Traseu Inginerie Software in FMI

